

Higher-Order Temporal Structure

Three-step and four-step forward/reverse analysis beyond net directional displacement

Standalone Handbook

A scientific, mathematical, biophysical, Python, machine-learning, validation, and reproducibility record of Phase 8B.

VALIDATED PHASE SNAPSHOT

Item	Validated state
Executed notebook	notebooks/08B_higher_order_temporal_structure_MSC01.ipynb
Notebook size	117 cells; 449,175 bytes; 0 saved error outputs
Primary design	3-step analysis; conditional 4-step extension
Tracking sources	CTC reference primary; TrackMate sensitivity
Representations	raw and co-moving
Models	grouped Logistic Regression; fixed RBF-SVM
Null	200 paired orientation-randomization replicates
Integrated analyses	16 pre-specified combinations
Central result	8/8 raw net signals above null; 0/16 internal-odd; 0/16 higher-order positive
Automated tests	52 passed after Phase 8B implementation
Final science commit	0628f8b - Complete Phase 8B higher-order temporal analysis

Prepared as a standalone continuation companion to the Phase 0-4 Master Handbook and the Phase 5, Phase 6, Phase 7, and Phase 8A records.

Repository: <https://github.com/AbolDpirate/cell-irreversibility>

Contents

Contents	2
1. Document purpose, scope, and evidence standard	4
2. Prior scientific context and the Phase 8B question	5
2.1 What Phase 7 had already established	5
2.2 Scientific question	5
2.3 Why a positive AUC alone was not enough	5
3. Pre-specification before outcome inspection	6
3.1 Why the support gate matters	6
3.2 Why CTC and TrackMate are not biological replicates	6
4. Data inputs, temporal support, and the four-step gate	7
4.1 Exact continuity rule	7
5. Generic exact k-step sequence infrastructure	8
5.1 Generic builder	8
5.2 Generic reversal	8
5.3 Backward-compatibility regression	8
6. Three-step time reversal and parity coordinates	9
6.1 Exact reversal	9
6.2 Parity decomposition	9
6.3 Invertibility	9
6.4 Synthetic and real-data validation	9
7. Co-moving reconstruction regression and pairing integrity	10
8. Three-step feature spaces and grouped validation	11
8.1 Balanced paired classification dataset	11
8.2 GroupKFold audit	11
9. Paired orientation-randomization null	12
9.1 Primary higher-order success rule	12
10. Three-step logistic results	13
10.1 Primary CTC co-moving result	13
10.2 CTC raw supporting result	13
10.3 TrackMate tracking-source sensitivity	13
11. Three-step nonlinear RBF sensitivity	14
12. Four-step time reversal and parity coordinates	15
12.1 Exact reversal	15
12.2 Parity coordinates	15
12.3 Inverse transform	15
13. Four-step real-data structural validation	16
13.1 Four-step feature spaces	16

13.2 Four-step grouped fold audit	16
14. Four-step logistic results	17
14.1 Primary four-step CTC co-moving result	17
14.2 Raw results	17
15. Four-step nonlinear RBF sensitivity	18
16. Integrated 16-analysis synthesis	19
16.1 Complete numerical ledger	20
17. Scientific interpretation	21
17.1 What Phase 8B supports	21
17.2 What Phase 8B does not support	21
17.3 Why the negative result is useful	21
18. Python, mathematics, statistics, and ML concepts reinforced	22
19. Reproducibility, testing, and Git chronology	23
19.1 Test progression	23
19.2 Final notebook reproducibility	23
19.3 Git chronology	23
20. Mistakes, debugging, and corrections	24
21. Notebook execution ledger	25
22. Reusable code added or expanded in Phase 8B	26
22.1 src/sequences.py	26
22.2 Tests	26
22.3 Notebook-only helpers	26
23. Scientific limitations and open questions	27
24. Handoff to Phase 9 - nonequilibrium thermodynamic interpretation	28
24.1 Phase 9 scientific tasks	28
24.2 Fresh-session starting state	28
25. Glossary	29
Appendix A. Key code patterns	30
A.1 Three-step parity decomposition	30
A.2 Four-step parity decomposition	30
A.3 Fixed primary model	30
A.4 Fixed nonlinear sensitivity model	30
A.5 Empirical upper-tail p-value convention	30
Appendix B. Phase 8B reproducibility checklist	31
Appendix C. Final handoff prompt	32

Major sections intentionally begin on new pages. The executable notebook remains the exact source of code-cell order; this handbook preserves the scientific logic, validated outputs, decisions, corrections, and reusable code that define the phase.

1. Document purpose, scope, and evidence standard

Phase 8B is the higher-order temporal-structure phase of the finite Cell Motility Irreversibility project. Its purpose was not to search broadly for any classifier that could separate forward from reversed sequences. The narrower scientific question was whether longer ordered displacement sequences contain **time-direction information beyond net directional displacement**.

This handbook records the complete scientific decision chain: why three-step sequences were primary, why four-step sequences were conditional, how exact reversal was derived, how parity coordinates were constructed, why grouped cross-validation and paired orientation randomization were mandatory, what every major output showed, and why the negative higher-order result is scientifically informative rather than a failure.

Evidence item	What it establishes
Executed Notebook 08B	Authoritative analysis order, saved outputs, 117-cell final notebook, final integrated summary.
Pre-specified design commit 3cd6a20	Scientific rules were frozen before Phase 8B classifier outcomes were inspected.
Generic k-step and parity source code	Reusable exact sequence construction and exact 3-step/4-step reversal-parity algebra.
Automated tests	Synthetic invariants, backward compatibility, parity transformation, inversion and endpoint identities; final suite 52 passed.
Recorded terminal/Git outputs	Commit chronology, branch state, fresh-kernel rerun, zero saved errors.
Prior phase handbooks	Immediate context for the Phase 7 net-displacement finding and Phase 8A tracking-source validation.

Documentation boundary. Phase 8B is a disciplined exploratory analysis and educational/reproducibility artifact. It is not a publication-grade proof that the cell dynamics are reversible, and it does not measure physical entropy production.

Exact stopping point. The scientific analysis is complete. No five-step search, classifier zoo, feature fishing, AUC flipping, or hyperparameter tuning is allowed inside Phase 8B after observing the null higher-order result.

2. Prior scientific context and the Phase 8B question

2.1 What Phase 7 had already established

Phase 7 asked whether Phase 6 had thrown away too much information by reducing two-step sequences. Full two-step forward/reverse classification did detect a raw signal, but reversal decomposition showed that almost all linear information was reproduced by the two-frame net displacement $d1 + d2$. The reversal-even component was exactly chance, and the fixed RBF sensitivity model did not reveal convincing extra full-sequence information.

The immediate Phase 8B implication was therefore: change **temporal depth**, not merely the classifier. Longer sequences were the next controlled test of whether temporal memory exists beyond the two-step interval.

2.2 Scientific question

Primary question: after accounting for net directional displacement, do exact three-step sequences contain detectable forward-versus-reversed information?

Secondary question: if sample support is adequate under a rule fixed in advance, does a four-step extension reveal higher-order information that the three-step analysis misses?

2.3 Why a positive AUC alone was not enough

A forward/reverse classifier can be successful for a trivial geometric reason: the net displacement changes sign under reversal. Therefore, Phase 8B could not define success as $AUC > 0.5$. It had to test whether an internal time-odd coordinate or a complete sequence representation adds information **beyond net displacement**.

This distinction is the central conceptual contribution of the phase.

3. Pre-specification before outcome inspection

Phase 8B began with a Git-recorded analysis design. This was an internal pre-registration: the representation, models, nulls, sample gate, grouping, success rule, and stopping rule were committed before Phase 8B outcome statistics were inspected.

Design item	Frozen choice
Primary sequence length	3 displacement steps
Conditional secondary length	4 steps
4-step support gate	≥ 200 exact sequences AND ≥ 12 trajectory groups
Primary tracking source	CTC reference
Sensitivity tracking source	TrackMate
Representations	raw and co-moving
CV	3-fold GroupKFold by trajectory
Primary model	StandardScaler -> LogisticRegression(max_iter=2000)
Nonlinear sensitivity	StandardScaler -> RBF-SVM(C=1, gamma="scale")
Null	paired orientation randomization
Null replicates	200
Logistic null seed	2029
RBF null seed	2030
Primary higher-order contrast	odd_combined AUC - net_only AUC
Positive criterion	odd_combined above paired null AND odd-minus-net above paired contrast null
Forbidden adaptation	AUC flipping, classifier zoo, outcome-driven features, hyperparameter search

3.1 Why the support gate matters

A four-step sequence uses five consecutive observations and therefore loses support faster than a three-step sequence. The gate was fixed before counting the four-step sequences. This prevents the threshold from being lowered after seeing whether four-step analysis would be convenient or statistically favorable.

3.2 Why CTC and TrackMate are not biological replicates

Both tracking sources come from the same microscopy sequence. CTC is a tracking-reference reconstruction; TrackMate is an independently reconstructed tracking pipeline. Agreement tests robustness to trajectory reconstruction, not independent biological generalization.

4. Data inputs, temporal support, and the four-step gate

Source	Observations	Tracks before higher-order filtering
CTC reference	428	15
TrackMate raw	474	35
TrackMate co-moving	474	35

The first data-dependent computation was intentionally limited to support counting. No AUC, null, p-value, or irreversibility statistic was inspected at that stage.

Tracking source	Exact 3-step sequences	3-step groups	Exact 4-step sequences	4-step groups	4-step eligible?
CTC reference	383	14	369	14	Yes
TrackMate	357	28	329	26	Yes

Both sources passed the pre-specified four-step gate. The four-step analysis therefore remained in scope as the planned secondary extension.

4.1 Exact continuity rule

An n-step sequence requires positions at $t, t+1, \dots, t+n$ for the same track. Row adjacency is not sufficient. Missing intermediate frames invalidate the window. This is the higher-order analogue of the exact-lag rule established earlier in the project.

Exact continuity logic

```
# Conceptual rule
for start frame t:
  require same cell at t, t+1, ..., t+n
  only then construct d1, d2, ..., dn
```

5. Generic exact k-step sequence infrastructure

Before counting or classifying real higher-order sequences, reusable generic k-step utilities were added to `src/sequences.py` and protected with synthetic unit tests. Historical two-step functions were retained so Phase 6/7 definitions remained stable.

5.1 Generic builder

```
def build_k_step_sequences(tracks, n_steps):
    validate_tracks(tracks)
    one_frame_steps = compute_steps_for_tau(tracks, tau_frames=1)
    # Merge successive exact 1-frame steps on cell_id and shared frame.
    # Explicitly verify frame_start + position == observed frame.
    # Return ordered dx1,dy1,...,dxn,dyn plus span metadata.
```

The design builds arbitrary ordered sequences from validated exact one-frame steps rather than using positional shifts through a table. The final continuity assertions make the temporal rule executable, not merely descriptive.

5.2 Generic reversal

```
def reverse_k_step_sequences(sequence_array):
    step_vectors = sequence_array.reshape(n_sequences, -1, 2)
    reversed_vectors = -step_vectors[:, ::-1, :]
    return reversed_vectors.reshape(sequence_array.shape)
```

The reshape exposes each displacement as a 2-vector. Reversing the step axis and multiplying by -1 implements the exact path reversal for any k.

5.3 Backward-compatibility regression

The generic builder was tested at `n_steps=2` against the validated historical `build_two_step_sequences` implementation. Sequence counts, cell IDs, endpoint frames, and displacement arrays matched. This prevented Phase 8B refactoring from silently changing the meaning of earlier phases.

Test progression: 36 tests after Phase 8A -> 41 after generic k-step tests -> 42 after explicit historical two-step equivalence.

6. Three-step time reversal and parity coordinates

6.1 Exact reversal

$$\begin{aligned} S3 &= (d1, d2, d3) \\ R(S3) &= (-d3, -d2, -d1) \end{aligned}$$

6.2 Parity decomposition

$$\begin{aligned} n &= d1 + d2 + d3 \\ q &= d1 - 2*d2 + d3 \\ e &= d1 - d3 \end{aligned}$$

Under exact reversal, $n \rightarrow -n$, $q \rightarrow -q$, and $e \rightarrow e$. The net coordinate n is exactly the endpoint displacement $r(t+3)-r(t)$ by telescoping. The internal odd coordinate q depends on how displacement is distributed across the three ordered steps and is therefore the key candidate for time-odd internal structure.

6.3 Invertibility

$$\begin{aligned} d2 &= (n - q) / 3 \\ d1 &= (2*n + q + 3*e) / 6 \\ d3 &= (2*n + q - 3*e) / 6 \end{aligned}$$

Because the transformation is invertible, $[n,q,e]$ contains the same information as the original six-dimensional $[d1,d2,d3]$ sequence. The decomposition changes interpretability, not information content.

6.4 Synthetic and real-data validation

- Manual-value tests verified n , q , and e on hand-checkable arrays.
- Reversal tests verified $n_R=-n$, $q_R=-q$, $e_R=e$.
- Round-trip tests reconstructed the original six-dimensional sequence.
- A telescoping-identity test verified $n = r(t+3)-r(t)$.
- Real-data maximum round-trip errors were about $3.55e-15$ for CTC and $1.78e-15$ for TrackMate.

After these additions the automated suite reached 47 passing tests.

7. Co-moving reconstruction regression and pairing integrity

CTC reference trajectories did not need a separately committed co-moving CSV. Instead, the previously used frame-wise common-motion transformation was reconstructed and first regression-tested on TrackMate against the saved historical co-moving trajectory table.

```
one_frame_steps -> group by frame transition
common_dx = median(dx across cells)
common_dy = median(dy across cells)
cumulative correction = cumulative sum(common motion)
co-moving coordinate = raw coordinate - cumulative correction
```

Check	Result
TrackMate common-motion transitions	47
Max x reconstruction difference	5.684e-14
Max y reconstruction difference	2.842e-14
Historical TrackMate co-moving regression	PASSED
CTC co-moving observations/tracks	428 / 15
CTC raw/co-moving 3-step windows	Identical
TrackMate raw/co-moving 3-step windows	Identical

The ~1e-14 discrepancies are normal floating-point round-off. This check established that Phase 8B reused the historical co-moving definition rather than inventing a new correction.

8. Three-step feature spaces and grouped validation

Feature space	Definition	Dimension	Reversal role
net_only	n	2D	odd; net displacement
internal_odd_only	q	2D	odd; internal structure
even_only	e	2D	even negative control
odd_combined	[n,q]	4D	all linear odd coordinates
full_6d	[d1,d2,d3]	6D	complete physical sequence

The full model was fixed in the original displacement coordinates before AUC inspection. Although [d1,d2,d3] and [n,q,e] are linearly invertible, StandardScaler plus regularized classification is not invariant to an arbitrary change of coordinates.

8.1 Balanced paired classification dataset

Every observed forward sequence was paired with its exact reverse. Labels were fixed as forward=1 and reversed=0. AUC values below 0.5 remained below 0.5; class meaning was never flipped post hoc.

8.2 GroupKFold audit

Source	Forward sequences	Classifier examples	Groups	Test groups by fold	Overlap
CTC	383	766	14	4 / 5 / 5	0
TrackMate	357	714	28	9 / 9 / 10	0

Every fold preserved exact 50/50 class balance because each biological sequence contributes one forward and one reversed example. Grouping prevents related windows from the same trajectory from appearing in both training and test sets.

9. Paired orientation-randomization null

The null preserved the empirical sequence geometry and forward/reverse pairing. Within each replicate, every sequence pair was independently orientation-swapped with probability 0.5. The same swap mask was applied to all feature spaces being compared in that replicate.

```
for replicate in range(200):  
    swap_mask = RNG.random(n_sequences) < 0.5  
    use the SAME mask for net, internal-odd, odd-combined, and full  
    run grouped CV  
    store AUCs and paired AUC differences
```

This design calibrates both individual AUCs and the central model contrast. With 200 replicates, the minimum empirical upper-tail p-value is $1/201 = 0.004975$.

9.1 Primary higher-order success rule

The higher-order criterion required both: (1) `odd_combined` above its paired-null 95th percentile, and (2) `odd_combined - net_only` above its paired contrast-null 95th percentile. A strong net-direction signal alone therefore could not be mislabeled as higher-order temporal structure.

10. Three-step logistic results

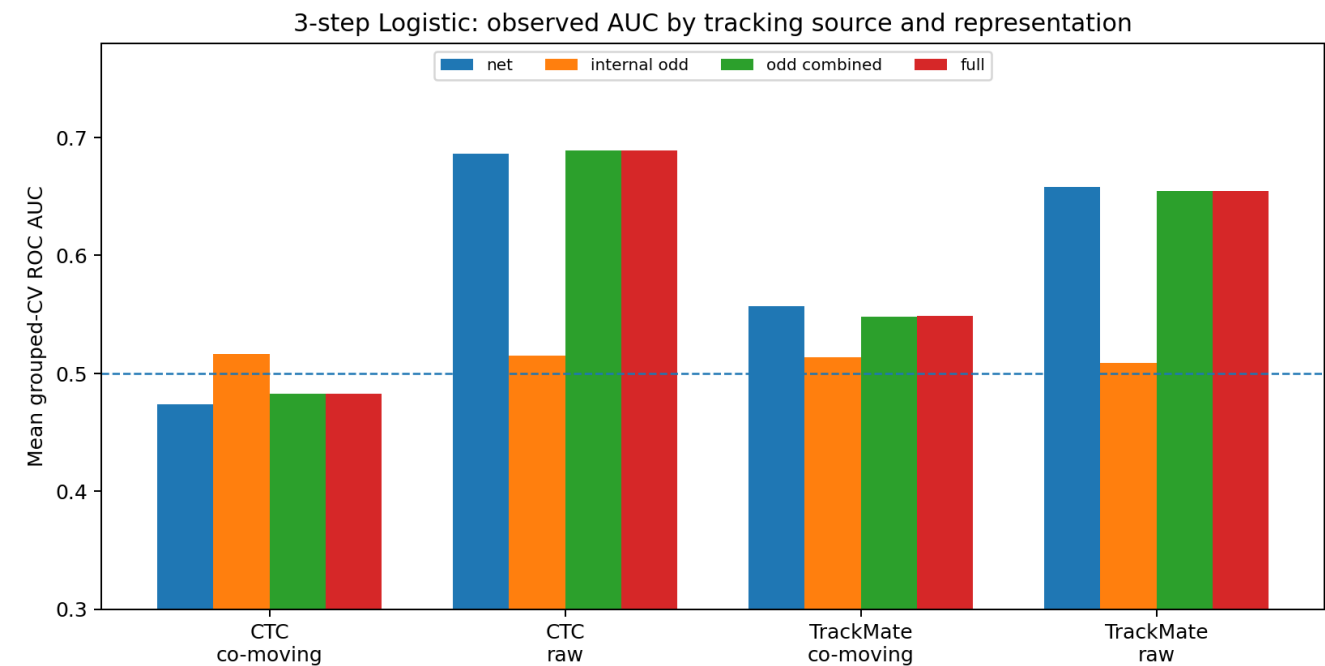


Figure 1. Three-step logistic mean grouped-CV ROC AUC. Raw trajectories show directional signal; co-moving representations do not show convincing above-null classification.

Source / repr.	net	internal odd	odd combined	full	Key contrast odd-net
CTC co-moving	0.474	0.516	0.483	0.483	+0.0086 (not above null)
CTC raw	0.687	0.515	0.689	0.689	+0.0025 (not above null)
TrackMate co-moving	0.557	0.514	0.548	0.549	-0.0090 (not above null)
TrackMate raw	0.658	0.509	0.655	0.655	-0.0032 (not above null)

10.1 Primary CTC co-moving result

None of the five feature spaces exceeded the paired-null 95th percentile. The pre-specified odd_combined-minus-net contrast was 0.008585, compared with null q95=0.053515 and empirical p=0.378109. The primary higher-order evidence criterion was therefore False.

10.2 CTC raw supporting result

Raw net_only, odd_combined, and full_6d all exceeded their paired nulls at the minimum empirical p=0.004975. However, internal_odd_only did not. The gain from odd_combined over net_only was only 0.002486 and was null-compatible (p about 0.468).

10.3 TrackMate tracking-source sensitivity

TrackMate reproduced the same qualitative result. Co-moving representations stayed below their null thresholds. Raw net, odd-combined, and full models were above null, but neither internal odd information nor odd-minus-net improvement was detected.

Three-step linear conclusion: raw forward/reverse information is detectable, but it is explained by net directional displacement rather than detectable internal ordering.

11. Three-step nonlinear RBF sensitivity

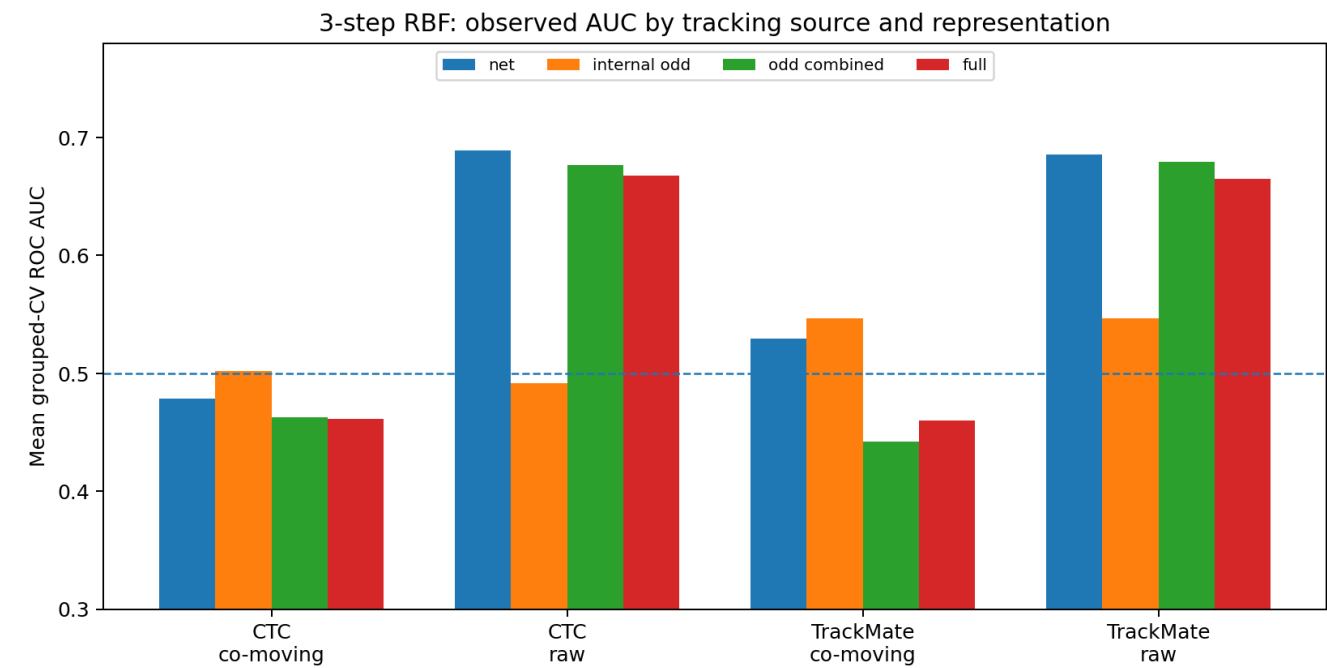


Figure 2. Three-step fixed RBF-SVM results. The nonlinear sensitivity model does not rescue a higher-order signal.

Source / repr.	net	internal odd	odd combined	full	odd-net
CTC co-moving	0.478	0.502	0.463	0.461	-0.0154
CTC raw	0.689	0.492	0.677	0.668	-0.0119
TrackMate co-moving	0.530	0.547	0.442	0.460	-0.0873
TrackMate raw	0.686	0.547	0.679	0.665	-0.0063

Raw net, odd-combined, and full models were again above null in CTC and TrackMate. Co-moving models remained null-calibrated negative. Crucially, every odd-minus-net contrast was null-compatible. The fixed nonlinear model therefore reinforced the logistic interpretation rather than changing it.

No classifier tuning followed this null result, in accordance with the pre-specified stopping rule.

12. Four-step time reversal and parity coordinates

12.1 Exact reversal

```
S4 = (d1, d2, d3, d4)
R(S4) = (-d4, -d3, -d2, -d1)
```

12.2 Parity coordinates

```
n4 = d1 + d2 + d3 + d4
q4 = d1 - d2 - d3 + d4
e41 = d1 - d4
e42 = d2 - d3
```

Under reversal, $n4$ and $q4$ change sign; $e41$ and $e42$ do not. The even subspace is now four-dimensional because it contains two two-dimensional invariant vectors.

12.3 Inverse transform

```
outer_sum = (n4 + q4) / 2
inner_sum = (n4 - q4) / 2
d1 = (outer_sum + e41) / 2
d4 = (outer_sum - e41) / 2
d2 = (inner_sum + e42) / 2
d3 = (inner_sum - e42) / 2
```

Five new tests checked manual values, parity under reversal, invertibility, the four-step endpoint telescoping identity, and invalid-shape handling. The full automated suite then reached 52 passing tests.

13. Four-step real-data structural validation

Check	CTC	TrackMate
Raw sequences	369	329
Co-moving sequences	369	329
Raw/co-moving window identity	True	True
Array shapes	(369, 8)	(329, 8)
Reversal involution	PASSED	PASSED
Max round-trip error raw	3.55e-15	6.66e-16
Max round-trip error co-moving	3.55e-15	8.88e-16

13.1 Four-step feature spaces

Feature space	Definition	Dimension
net_only	n4	2D
internal_odd_only	q4	2D
even_only	[e41,e42]	4D
odd_combined	[n4,q4]	4D
full_8d	[d1,d2,d3,d4]	8D

13.2 Four-step grouped fold audit

Source	Examples	Groups	Test groups by fold	Group overlap
CTC	738	14	4 / 5 / 5	0
TrackMate	658	26	8 / 9 / 9	0

No four-step AUC was inspected until the temporal windows, parity algebra, feature matrices, negative control, balanced datasets, and grouped folds had all passed their structural audits.

14. Four-step logistic results

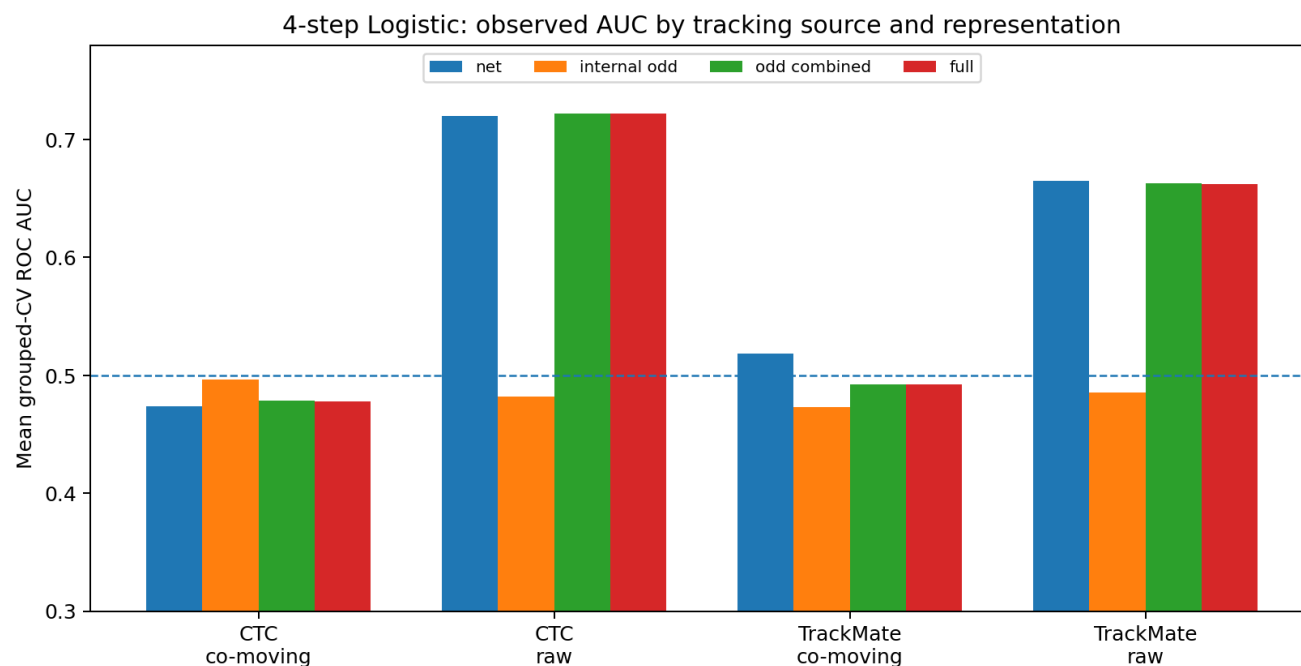


Figure 3. Four-step logistic AUCs. The four-step extension reproduces the three-step pattern rather than revealing hidden internal time order.

Source / repr.	net	internal odd	odd combined	full	odd-net
CTC co-moving	0.474	0.496	0.478	0.478	+0.0044
CTC raw	0.720	0.482	0.722	0.722	+0.0021
TrackMate co-moving	0.519	0.473	0.492	0.492	-0.0263
TrackMate raw	0.665	0.486	0.663	0.662	-0.0024

14.1 Primary four-step CTC co-moving result

No feature space exceeded the paired-null 95th percentile. The odd-minus-net contrast was 0.004368 versus null $q_{95}=0.061151$ with empirical $p=0.368159$. The four-step higher-order criterion was False.

14.2 Raw results

CTC raw and TrackMate raw again showed above-null net, odd-combined, and full classifier performance. Yet internal_odd_only remained null, and the additional performance over net displacement was effectively zero and not above the paired contrast null.

Four-step linear conclusion: increasing sequence length did not expose internal temporal ordering that was hidden at three steps.

15. Four-step nonlinear RBF sensitivity

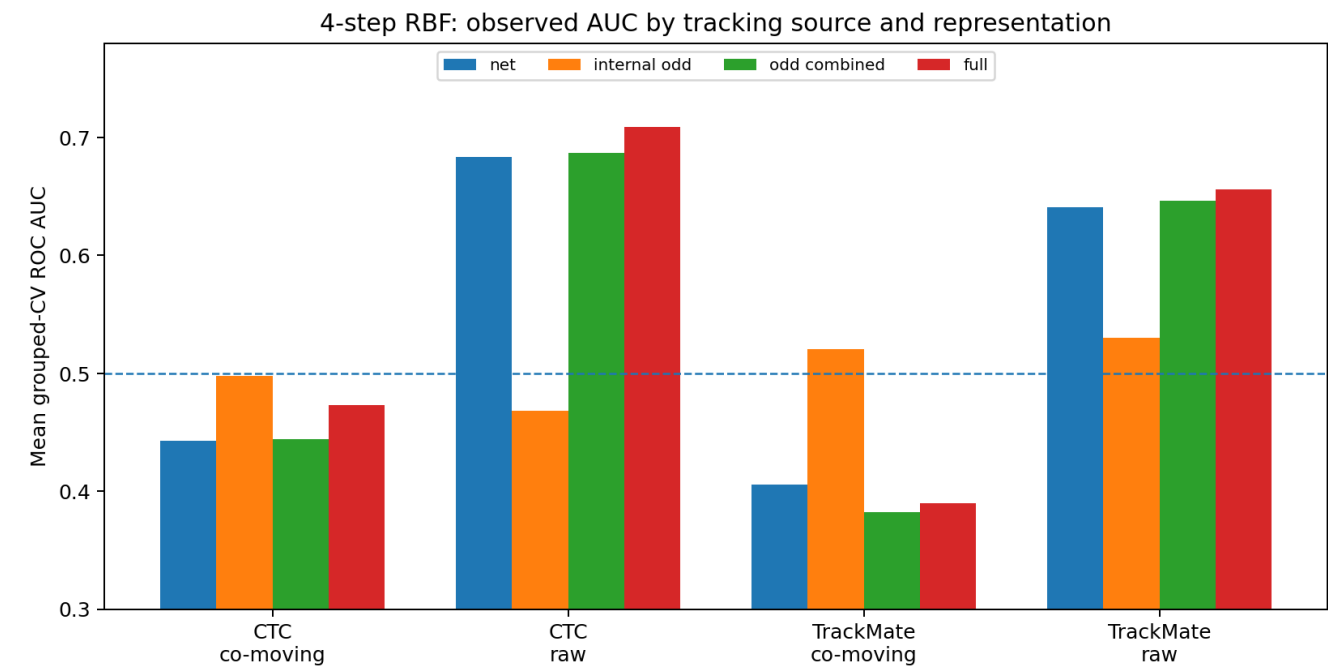


Figure 4. Four-step fixed RBF-SVM results. Raw directional signal persists; co-moving higher-order evidence remains absent.

Source / repr.	net	internal odd	odd combined	full	odd-net / full-net
CTC co-moving	0.443	0.498	0.444	0.473	+0.0014 / +0.0308
CTC raw	0.684	0.468	0.687	0.709	+0.0033 / +0.0254
TrackMate co-moving	0.406	0.521	0.382	0.390	-0.0236 / -0.0160
TrackMate raw	0.641	0.530	0.647	0.656	+0.0054 / +0.0150

No co-moving feature space was above null. Raw CTC and TrackMate again showed net, odd-combined, and full signals, but all pre-specified contrasts beyond net displacement were null-compatible. The RBF result therefore did not change the scientific interpretation.

16. Integrated 16-analysis synthesis

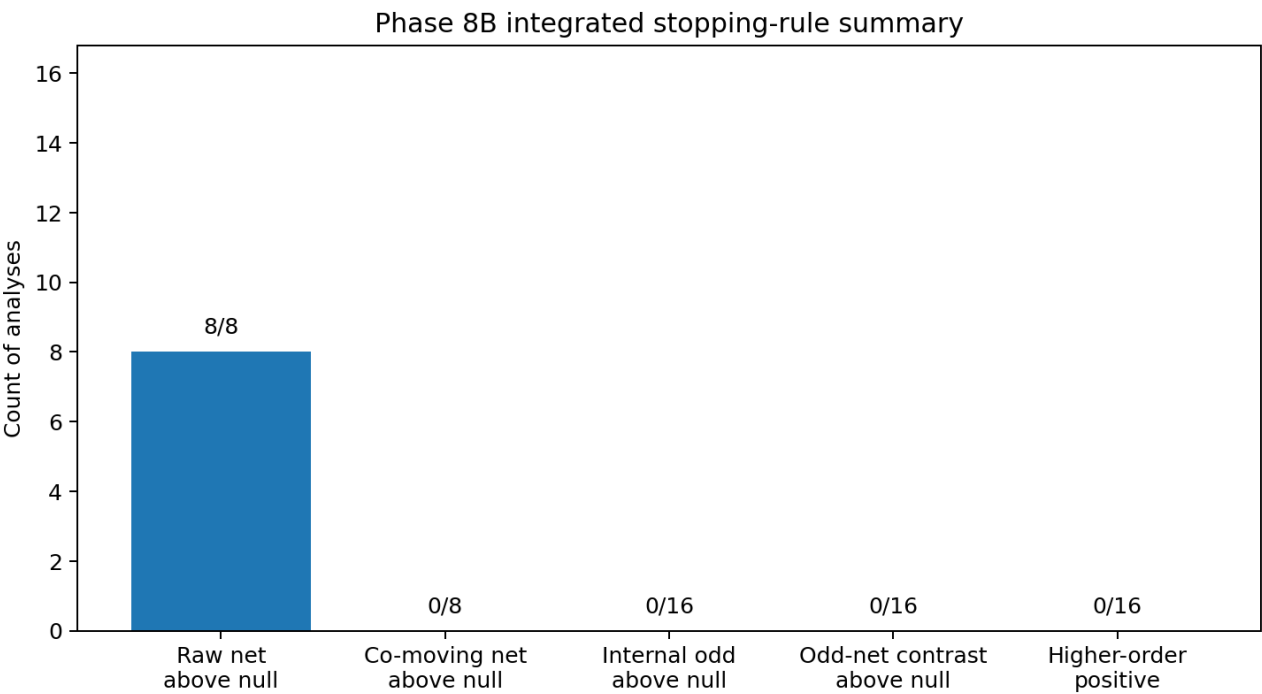


Figure 5. Integrated Phase 8B stopping-rule counts. Higher-order evidence is absent in every pre-specified analysis.

Diagnostic	Result
Integrated analyses	16
Raw net-displacement signal above null	8 / 8
Co-moving net-displacement signal above null	0 / 8
Internal reversal-odd signal above null	0 / 16
Odd-minus-net contrast above null	0 / 16
Higher-order positive analyses	0 / 16

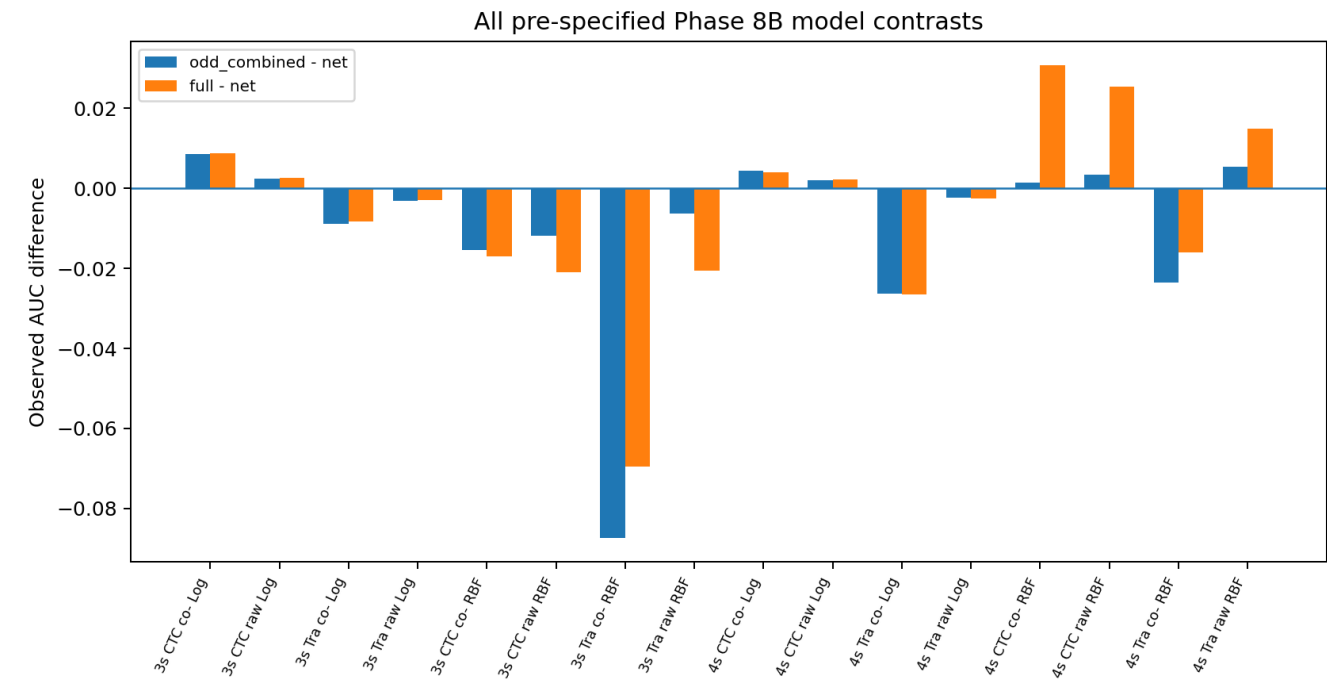


Figure 6. Observed model contrasts across all 16 analyses. Some differences are numerically positive or negative, but none crossed its paired null threshold.

16.1 Complete numerical ledger

Steps	Source	Repr.	Model	Net	Internal	Odd comb.	Full	Net>null	Odd-net
3	CTC	co-moving	Logistic	0.474	0.516	0.483	0.483	No	+0.009
3	CTC	raw	Logistic	0.687	0.515	0.689	0.689	Yes	+0.002
3	TrackMate	co-moving	Logistic	0.557	0.514	0.548	0.549	No	-0.009
3	TrackMate	raw	Logistic	0.658	0.509	0.655	0.655	Yes	-0.003
3	CTC	co-moving	RBF	0.478	0.502	0.463	0.461	No	-0.015
3	CTC	raw	RBF	0.689	0.492	0.677	0.668	Yes	-0.012
3	TrackMate	co-moving	RBF	0.530	0.547	0.442	0.460	No	-0.087
3	TrackMate	raw	RBF	0.686	0.547	0.679	0.665	Yes	-0.006
4	CTC	co-moving	Logistic	0.474	0.496	0.478	0.478	No	+0.004
4	CTC	raw	Logistic	0.720	0.482	0.722	0.722	Yes	+0.002
4	TrackMate	co-moving	Logistic	0.519	0.473	0.492	0.492	No	-0.026
4	TrackMate	raw	Logistic	0.665	0.486	0.663	0.662	Yes	-0.002
4	CTC	co-moving	RBF	0.443	0.498	0.444	0.473	No	+0.001
4	CTC	raw	RBF	0.684	0.468	0.687	0.709	Yes	+0.003
4	TrackMate	co-moving	RBF	0.406	0.521	0.382	0.390	No	-0.024
4	TrackMate	raw	RBF	0.641	0.530	0.647	0.656	Yes	+0.005

The apparent simplicity of the final count table is earned by extensive controls: exact continuity, algebraic parity, strict pairing, grouped folds, pre-fixed model classes, and paired nulls. The result is not based on a single failed classifier.

17. Scientific interpretation

17.1 What Phase 8B supports

- Raw trajectories consistently contain forward/reverse information across 3-step and 4-step sequence lengths, both tracking reconstructions, and both tested classifiers.
- The raw information is already present in net displacement; adding internal reversal-odd structure does not produce detectable improvement.
- The internal odd coordinates q and q_4 are never above their paired nulls across the 16 integrated analyses.
- After common-motion subtraction, neither tracking reconstruction provides convincing higher-order forward/reverse evidence.
- The qualitative conclusion is robust to tracking source and to the fixed nonlinear sensitivity model.

Central conclusion: extending ordered trajectory analysis from two steps to three and four steps did not reveal detectable higher-order time-direction information beyond net directional displacement at the available data scale.

17.2 What Phase 8B does not support

- It does not prove microscopic time reversibility.
- It does not establish detailed balance.
- It does not show that physical entropy production is zero.
- It does not rule out all longer-timescale memory or all possible observables.
- It does not justify calling classifier AUC a thermodynamic arrow of time.
- It does not identify the physical origin of the common population motion.

17.3 Why the negative result is useful

A negative higher-order result closes an important ambiguity from Phase 7. If the two-step analysis had missed internal temporal order simply because the sequence was too short, three-step or four-step parity coordinates might have revealed it. They did not. The evidence therefore points consistently toward net directional translation as the dominant source of raw forward/reverse separability in MSC01 under the tested scales.

18. Python, mathematics, statistics, and ML concepts reinforced

Concept	Phase 8B use
Lists / tuples	Fixed ordered groups of sequence components and return values.
Dictionaries	Named feature-space collections and nested result bundles.
NumPy reshape	Convert flat 2k arrays into sequence x step x coordinate tensors.
np.column_stack	Assemble parity feature matrices while preserving row pairing.
pandas merge	Build exact k-step windows by shared cell/frame endpoints.
DataFrame summaries	Create reusable observed/null/contrast result tables.
GroupKFold	Protect against trajectory leakage across train/test sets.
Pipeline	Fit scaling only on training data before logistic/RBF estimation.
ROC AUC	Measure forward/reverse ranking without changing class meaning post hoc.
Balanced accuracy	Secondary classification diagnostic under exactly balanced labels.
Permutation/randomization logic	Generate a pair-preserving orientation null.
Empirical p-values	Upper-tail calibration with finite 200-replicate nulls.
Involution	$R(R(S))=S$ as a code-level scientific invariant.
Linear invertibility	Parity coordinates preserve the full sequence information.
Telescoping sum	Net odd coordinate equals endpoint displacement.
Pre-specification	Separate methodological choice from observed outcome.

19. Reproducibility, testing, and Git chronology

19.1 Test progression

Checkpoint	Tests
Phase 8A final baseline	36 passed
Generic k-step infrastructure	41 passed
Historical two-step equivalence	42 passed
Three-step parity suite	46 passed
Three-step endpoint identity	47 passed
Four-step parity suite	52 passed
Final Phase 8B repository validation	52 passed in about 3 s

19.2 Final notebook reproducibility

Item	Final validated value
Notebook cells	117
Executed code cells after fresh Run All	92
Saved error outputs	0
Notebook file size	449,175 bytes
Fresh-kernel Run All	completed
Integrated validation	PASSED
Scientific status output	COMPLETE

19.3 Git chronology

Commit	Meaning
3cd6a20	Pre-specify Phase 8B higher-order temporal analysis
e568e50	Add generic exact k-step sequence utilities
9d33342	Record Phase 8B sequence-support audit
fef0d24	Add three-step reversal-parity coordinates
70984d8	Lock Phase 8B three-step feature and fold design
62a733a	Record primary Phase 8B three-step logistic result
658d699	Record CTC three-step logistic analyses
1beb2ed	Complete three-step logistic tracking-source analysis
0eef8bc	Complete Phase 8B three-step analysis
f689217	Add four-step reversal-parity coordinates
f75b5f7	Lock Phase 8B four-step feature and fold design
98bcde3	Complete Phase 8B four-step logistic analysis
0628f8b	Complete Phase 8B higher-order temporal analysis

The chronology is scientifically meaningful: design -> infrastructure -> support audit -> algebra -> pre-classification fold lock -> outcomes. This makes it possible to audit which choices were genuinely made before result inspection.

20. Mistakes, debugging, and corrections

Issue	Diagnosis	Resolution / lesson
Four new parity tests failed with NameError	New functions existed in src/sequences.py but were not imported into notebook	add the new src_sequences.py to terminal import passed; 46 tests then passed.
Notebook Cell 11 ImportError after source	Jupyter kernel had cached an older src.sequences module.	Restart kernel and rerun; no scientific code change was needed.
Markdown/LaTeX rendering looked broken	Nested chat code fences / delimiters rendered poorly in Jupyter	Use one clean Markdown block and $...$ display math.
Concern about blank-line git warnings in Jupyter	Disproportionate whitespace warnings were not scientific failures.	Do not over-prioritize non-substantive formatting after git diff --check is clean.
Potential temptation to interpret AUC below 0.5 as post-hoc class flipping.	Visualize post-hoc	AUC sign/direction remained as observed; no post-hoc inversion.
Potential temptation to add models after results	Would violate stopping rule and inflate discovery flexibility.	No classifier zoo, feature search, 5-step search, or tuning.

21. Notebook execution ledger

The final executed notebook contains 117 cells: 92 code cells and 25 Markdown cells. The executable .ipynb file is the authoritative 1:1 record. The ledger below maps the complete notebook into auditable scientific blocks while avoiding a misleading reconstruction of every Jupyter split/merge that occurred during interactive work.

Notebook block	Primary purpose	Validated outputs / decisions
Cells 1-9	Load sources; standardize schema; support audit; record 4-step CTC	CTC 428/15; TrackMate 474/35; 3-step 383/357; 4-step 369/329; both sources eligible.
Cells 10-22	3-step arrays, exact reversal, parity algebra, co-moving regression	(338,8)/(357,8) windows PASSED; round-trip ~1e-15; TrackMate historical co-moving reconstructed to ~1e-14;
Cells 23-35	Fix 3-step feature spaces; build balanced classifier sets; grouped null studies	Fixed feature spaces fixed; even control identical; CTC folds 4/5/5 test groups; TrackMate 9/9/10; zero leakage.
Cells 36-46	Primary CTC co-moving logistic and paired null	Primary criterion False; odd-net +0.008585, p~0.378; no feature above null.
Cells 47-60	CTC raw and TrackMate logistic sensitivity; linear synthesis	Raw net/odd/full above null; internal odd and odd-net contrasts null-compatible across both sources.
Cells 61-68	Fixed 3-step RBF sensitivity	Same qualitative conclusion as logistic; three-step analysis closed.
Cells 69-86	4-step exact windows, parity, feature spaces and grouped null	CTC / 329 TrackMate; (369,8)/(329,8); round-trip ~1e-15; CTC folds 4/5/5; TrackMate 8/9/9.
Cells 87-100	Four-step logistic analyses and cross-source synthesis	CTC co-moving primary criterion False; raw net/odd/full above null; all odd-net contrasts null-compatible.
Cells 101-106	Fixed four-step RBF sensitivity	No higher-order rescue; even controls PASSED; final inferential block complete.
Cells 107-117	Integrated 16-analysis table, stopping-rule audit, final scientific conclusions and complete messages	Raw net/odd/full above null; odd 0/16; odd-net 0/16; higher-order positive 0/16; computational validation

22. Reusable code added or expanded in Phase 8B

22.1 `src/sequences.py`

- Generic n-step validation helper.
- `build_k_step_sequences` for exact arbitrary sequence length.
- `get_k_step_sequence_array` for ordered 2k-dimensional arrays.
- `reverse_k_step_sequences` for exact path reversal.
- `decompose_three_step_parity` and `reconstruct_three_step_from_parity`.
- `decompose_four_step_parity` and `reconstruct_four_step_from_parity`.

22.2 Tests

- Exact continuity and missing-frame rejection.
- Historical two-step backward compatibility.
- Reversal involution.
- Three-step and four-step manual parity values.
- Odd/even transformation laws under reversal.
- Invertibility / round-trip reconstruction.
- Endpoint telescoping identities.
- Shape and invalid-input validation.

22.3 Notebook-only helpers

Model-analysis helpers that mainly organize Phase 8B reporting remained notebook-local: feature-space dictionaries, grouped-fold audit summaries, logistic/RBF analysis wrappers, and integrated result-table builders. This avoids moving phase-specific orchestration into reusable source modules prematurely.

23. Scientific limitations and open questions

- The biological sample is one microscopy sequence; CTC and TrackMate are not independent biological datasets.
- Higher-order windows overlap heavily within trajectories, which is why grouping by trajectory is essential but does not create independent biological replicates.
- Only three-step and four-step sequences were tested; the project intentionally stops there to avoid result-driven sequence-length search.
- The co-moving transformation removes frame-wise median population motion, but the physical origin of that component remains unresolved.
- Classifier non-detection can result from limited sample size, weak signal, inappropriate observable, or genuine absence of detectable structure; it does not prove reversibility.
- The paired orientation null calibrates the classification problem, not thermodynamic entropy production.
- RBF-SVM was a fixed sensitivity model, not a comprehensive nonlinear function search.

24. Handoff to Phase 9 - nonequilibrium thermodynamic interpretation

Phase 9 begins after Phase 8B is merged and frozen. It must not relabel the current AUCs or displacement KL divergences as physical entropy production. The next phase should instead build the thermodynamic bridge explicitly and test it first on controlled processes where reversibility and dissipation structure are known.

24.1 Phase 9 scientific tasks

- Define forward and reversed path probabilities and their log-ratio carefully.
- Distinguish statistical path-space asymmetry from stochastic-thermodynamic entropy production.
- Construct reversible and irreversible synthetic controls with known qualitative behavior.
- Study finite-sample bias and coarse-graining effects.
- Determine when KL/path-probability quantities are legitimate entropy-production estimators or lower bounds.
- Only then apply defensible quantities to MSC01, with assumptions stated explicitly.
- Treat entropy-production rate and Thermodynamic Uncertainty Relation analyses as assumption-dependent, not automatic follow-ons.

24.2 Fresh-session starting state

Repository state after Phase 8B freeze:

- Phase 8B notebook: 117 cells, fresh-kernel Run All, 0 saved errors
- Automated tests: 52 passed
- Integrated higher-order positives: 0 / 16
- Raw net signals: 8 / 8
- Co-moving net signals: 0 / 8
- Do not reopen Phase 8B to force a positive result
- Start Phase 9 by pre-specifying the thermodynamic interpretation boundary

Phase 9 must proceed even though the experimental higher-order result is weak/null. The pedagogical and scientific goal is to learn exactly what thermodynamic quantities can and cannot be inferred from coarse trajectory data.

25. Glossary

Term	Meaning in this project
Exact k-step sequence	k consecutive one-frame displacement vectors requiring k+1 consecutive observations for the same track.
Time reversal	Reverse step order and negate every displacement vector.
Reversal-odd coordinate	Feature that changes sign under exact time reversal.
Reversal-even coordinate	Feature unchanged by exact time reversal.
Net displacement	Endpoint displacement across the full sequence; a reversal-odd but not inherently higher-order quantity.
Internal odd coordinate	Time-odd coordinate designed to depend on internal distribution/order of motion rather than net translation alone.
Co-moving representation	Coordinates after subtracting cumulative frame-wise median population motion.
GroupKFold	Cross-validation that assigns whole trajectories, rather than individual windows, to train/test folds.
Paired orientation null	Randomly swaps forward/reverse orientation within each exact pair while preserving geometry and pairing.
Higher-order evidence	Pre-specified joint criterion requiring odd-combined above null and odd-minus-net contrast above null.
Null-calibrated non-detection	Observed statistic does not exceed the chosen paired-null reference; not proof that the true signal is zero.
Tracking-source robustness	Qualitative agreement between CTC and TrackMate reconstructions of the same microscopy sequence.
Entropy production	Thermodynamic quantity deferred to Phase 9; not synonymous with AUC or generic KL asymmetry.

Appendix A. Key code patterns

A.1 Three-step parity decomposition

```
def decompose_three_step_parity(sequence_array):
    d = sequence_array.reshape(sequence_array.shape[0], 3, 2)
    d1, d2, d3 = d[:,0,:], d[:,1,:], d[:,2,:]
    net_odd = d1 + d2 + d3
    internal_odd = d1 - 2.0*d2 + d3
    reversal_even = d1 - d3
    return net_odd, internal_odd, reversal_even
```

A.2 Four-step parity decomposition

```
def decompose_four_step_parity(sequence_array):
    d = sequence_array.reshape(sequence_array.shape[0], 4, 2)
    d1,d2,d3,d4 = d[:,0,:],d[:,1,:],d[:,2,:],d[:,3,:]
    n4 = d1 + d2 + d3 + d4
    q4 = d1 - d2 - d3 + d4
    e41 = d1 - d4
    e42 = d2 - d3
    return n4, q4, e41, e42
```

A.3 Fixed primary model

```
Pipeline([
    ("scaler", StandardScaler()),
    ("classifier", LogisticRegression(max_iter=2000)),
])
```

A.4 Fixed nonlinear sensitivity model

```
Pipeline([
    ("scaler", StandardScaler()),
    ("classifier", SVC(kernel="rbf", C=1.0, gamma="scale")),
])
```

A.5 Empirical upper-tail p-value convention

```
p_emp = (1 + count(null_stat >= observed_stat)) / (1 + n_null)
# n_null = 200 -> minimum p = 1/201 = 0.004975
```

Appendix B. Phase 8B reproducibility checklist

Check	Final status
Pre-specified analysis document exists	Yes
Four-step threshold fixed before support counts	Yes
Generic k-step functions unit-tested	Yes
Historical two-step compatibility tested	Yes
Three-step parity invertibility tested	Yes
Four-step parity invertibility tested	Yes
Co-moving definition regression-checked against historical TrackMate	Yes
Raw/co-moving window identity checked	Yes
Even-only negative controls exact	Yes
Group leakage audits	Passed
Class balance in folds	0.5 / 0.5
Paired null masks shared across compared feature spaces	Yes
Post-hoc AUC flipping	No
Classifier tuning	No
Sequence lengths beyond 4 after results	No
Fresh-kernel Run All	Passed
Saved notebook error outputs	0
Automated tests	52 passed
Final science commit	0628f8b

Appendix C. Final handoff prompt

The following compact handoff can be used to start a fresh Phase 9 chat without reopening settled Phase 8B work.

We are continuing the finite Cell Motility Irreversibility project.
Treat the Master Handbook plus the Phase 5-8B standalone records as authoritative.

Validated Phase 8B state:

- Notebook 08B has 117 cells, fresh-kernel Run All, 0 saved errors.
- 52 automated tests pass.
- Three-step primary and conditional four-step secondary analyses are complete.
- CTC reference is primary; TrackMate is a tracking-source sensitivity reconstruction.
- Raw/co-moving, Logistic/RBF, and paired orientation-randomization nulls were all pre-specified.
- Integrated result: raw net above null 8/8; co-moving net 0/8; internal odd 0/16; odd-minus-net 0/16; higher-order positive 0/16.
- Interpretation: no detectable higher-order time-direction information beyond net displacement at the available data scale.
- This is NOT proof of time reversibility and NOT zero entropy production.
- Do not reopen Phase 8B to test more sequence lengths, features, classifiers, or tuned models.

Next: Phase 9 - nonequilibrium thermodynamic interpretation.

First pre-specify the thermodynamic boundary. Build reversible/irreversible synthetic controls, define forward/reversed path probabilities.